

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 5 : Session 8 : SLO 1 | ARM Pipeline, Registers and Instructions

I. Fill in the Blanks

1. Fetch is a 3-stage pipeline is Fetch, Decode, Execute.
2. Stack Pointer (SP) R13 points to the top of the stack.
3. BL (Branch with Link) calls a subroutine, saving the return address in R14 (LR).

II. True or False

1. TRUE pipelining overlaps instruction stages in ARM processors.
2. FALSE R14 is the Link Register; R15 is the Program Counter.
3. TRUE ADDEQ executes the ADD only when the Z (equal) condition flag is set.

III. Match the Following

1. R13 is 3. Stack Pointer.
2. R14 is 1. Link Register.
3. LDR is 2. Load instruction.

IV. Numerical Answers

1. 3 is three pipeline stages (Fetch, Decode, Execute).
2. 2 is one LDR plus one STR per iteration = 2 memory operations.
3. 13 is R0 through R12 are the general-purpose registers.